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# PAPER_1103: LQG Buoyancy Sector Lagrangian Variation with Spin-Foam Vertex Coupling

## Abstract

We construct the Lagrangian for the LQG buoyancy sector, coupling the spin-foam vertex amplitude $A_v(j, i_e)$ to the UQFF buoyancy scalar field $\phi_{\text{LQG}}$ and the 1.25 THz phonon mode. The Euler-Lagrange equation $\delta S/\delta\phi_{\text{LQG}} = 0$ yields an equation of motion with spin-foam source terms and phonon-induced effective mass corrections.

## 1. Lagrangian Density

$$\mathcal{L}_{\text{LQG}} = \tfrac{1}{2}(\partial_\mu \phi_{\text{LQG}})^2 - V(\phi_{\text{LQG}}) + \lambda_{\text{spin}} \sum_v A_v(j, i_e) \cdot \phi_{\text{LQG}} + g_{\text{phonon}}\, \Phi_{1.25\,\text{THz}}\, \phi_{\text{LQG}}^2$$

### Terms

| Term | Physical origin |
|---|---|
| $\frac{1}{2}(\partial_\mu \phi)^2$ | Kinetic energy of the buoyancy field |
| $V(\phi) = \frac{1}{2}\omega_\phi^2 \phi^2$ | Harmonic potential, $\omega_\phi = m_\phi c^2/\hbar$ |
| $\lambda_{\text{spin}} \sum_v A_v \cdot \phi$ | Spin-foam vertex coupling (EPRL/FK) |
| $g_{\text{phonon}} \Phi \phi^2$ | Phonon-buoyancy self-interaction |

## 2. Spin-Foam Vertex Amplitude

Using the Ponzano-Regge/EPRL approximation:

$$A_v(j) = (2j + 1) \cdot \exp(-\gamma j)$$

where $\gamma = 0.2375$ is the Barbero-Immirzi parameter. The sum over $N_v$ vertices:

$$\sum_v A_v = N_v \cdot (2j_{\text{avg}} + 1) \cdot \exp(-\gamma j_{\text{avg}})$$

## 3. Equation of Motion

Varying $\delta S/\delta\phi_{\text{LQG}} = 0$:

$$\Box\phi_{\text{LQG}} + \omega_\phi^2 \phi_{\text{LQG}} = \lambda_{\text{spin}} \sum_v A_v + 2g_{\text{phonon}}\, \Phi\, \phi_{\text{LQG}}$$

This is a **Klein-Gordon equation with spin-foam source and phonon-modified mass**:

$$\Box\phi + m_{\text{eff}}^2\phi = J_{\text{spin-foam}}$$

where:

$$m_{\text{eff}}^2 = \omega_\phi^2 - 2g_{\text{phonon}}\Phi$$

$$J_{\text{spin-foam}} = \lambda_{\text{spin}} \sum_v A_v(j, i_e)$$

## 4. Tachyonic Instability Condition

If $2g_{\text{phonon}}\Phi > \omega_\phi^2$, the effective mass squared becomes negative, signaling a tachyonic instability. This triggers spontaneous symmetry breaking in the buoyancy sector, with the new vacuum:

$$\langle\phi\rangle = \pm\sqrt{\frac{J_{\text{spin-foam}}}{m_{\text{eff}}^2}}$$

## 5. Static Equilibrium

At static equilibrium ($\Box\phi = 0$), the balance condition:

$$\omega_\phi^2\phi_0 = \lambda_{\text{spin}} \sum_v A_v + 2g_{\text{phonon}}\Phi\phi_0$$

The equation of motion residual $R$:

$$R = V'(\phi_0) - J_{\text{spin-foam}} - 2g_{\text{phonon}}\Phi\phi_0$$

with balance ratio $|R|/|V'(\phi_0)|$ indicating proximity to equilibrium.

## 6. Implementation

Calculator: `LQGBuoyancySectorLagrangianVariationCalculator` in CondensedPhysics.py

- Inputs: field amplitude $\phi_0$, mass $m_\phi$, couplings $\lambda_{\text{spin}}$ and $g_{\text{phonon}}$, vertex count $N_v$, average spin $j_{\text{avg}}$
- Outputs: Lagrangian density, spin-foam and phonon source terms, equation of motion residual, effective mass, tachyonic flag

## 7. Conclusion

The LQG buoyancy sector Lagrangian provides the variational framework for deriving the dynamics of the UQFF buoyancy field in the spin-foam formalism. The phonon-induced effective mass correction and the spin-foam source term together determine the equilibrium configuration and stability of the buoyancy sector.

## References


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